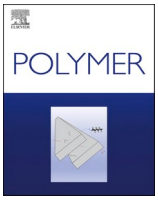
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Unlocking sequence distribution in ethylene/vinylcyclohexane copolymers by ^{13}C NMR: From microstructure to thermal properties

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ABSTRACT

Ethylene/Vinylcyclohexane (E/V) copolymers are attractive materials combining polyethylene-like processability with enhanced thermal properties. Until now, establishing quantitative relationships between chain microstructure and material properties has been hampered by the incomplete assignment of the ^{13}C NMR spectrum for samples with high V incorporation and consecutive V units, now accessible with certain non-metallocene catalysts. In this work, we address this limitation by means of natural-abundance 2D ^{13}C–^{13}C INADEQUATE NMR experiments, enabling a comprehensive assignment of the ^{13}C NMR spectrum of regioregular E/V copolymers. This, in turn, enables quantitative analysis of comonomer sequence distribution at the triad level and determination of monomer reactivity ratios for different catalyst systems. Copolymers with comparable V contents but different sequence distributions, from alternation-enriched to more blocky microstructures, exhibit significant differences in thermal behavior despite comparable compositions, highlighting the important role of sequence distribution in governing the thermal behavior of E/V copolymers.

1. Introduction

The pervasive success of polyolefin materials, now exceeding 50% by weight of the total polymer market [1–3], stems from the combination of a low cost and the fine control over chain microstructure ensured by coordination catalysts [4–8].

For ethylene/1-alkene copolymers, molecular catalysts provide the opportunity to tailor application properties, directly and indirectly (e. g., via crystallization or processing) [4,9], by modulating comonomer sequence distribution; for that, thorough ^{13}C NMR verification is key. A paradigmatic example is represented by flexible packaging films in which the crystallization ability of the polyethylene homosequences is limited by the introduction of higher 1-alkene comonomers in tunable amounts [10,11]. *Linear* higher 1-alkenes (namely 1-butene, 1-hexene and 1-octene) are mostly used due to a favorable cost-performance balance. Copolymers with comonomer contents below 5 mol%, indicatively, typically behave as semicrystalline thermoplastics (LLDPE-like materials). As comonomer incorporation increases, crystallinity progressively decreases, and the materials evolve toward plastomeric behavior (ca. 5–15 mol%), approaching thermoplastic elastomer characteristics at higher comonomer contents, where the copolymers become largely amorphous [12]. Accurate determinations of monomer reactivity ratios and product thereof are necessary to establish quantitative relationships between chain microstructure and physical material properties (that is thermal, mechanical, optical, barrier, and more) [13,14]. ^{13}C NMR spectral assignments are available in the literature since the 1980s, and analytical methods have long become routine [15].

Branched higher 1-alkenes (e. g. 4-methyl-1-pentene, 3-methyl-1-butene, vinylcyclohexane) are more effective than the linear homologs in disrupting PE crystallinity in films [13,16–18], and can determine peculiar material properties; until now, however, the high cost and a poor reactivity with metallocene catalysts were major drawbacks for practical utilizations. Recent advances in alternative olefin feedstocks (like, e. g. Fischer-Tropsch chemistry [19–21] or polyolefin waste upcycling [22,23]) together with the discovery of novel non-metallocene catalysts featuring enhanced incorporation ability [24,25], may reopen this area and stimulate renewed interest in unconventional polyolefin microstructures.

The ethylene/vinylcyclohexane (E/V) comonomer pair represents a particularly challenging case. Early copolymerization studies based on

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Group 4 metallocenes described copolymers with less than 5 mol% V units [26,27]. The later introduction of half-titanocene catalysts gave access to copolymers with much higher V contents (up to 30-40 mol%) but almost completely isolated V units, consistently with the observation that all such catalysts are substantially unable to homopolymerize V [28].

Only more recently, non-metallocene catalysts capable of incorporating large amounts of V and generating consecutive V sequences have been disclosed [24,25,29–34]. However, the lack of complete and reliable ^{13}C NMR assignments for such copolymers still hampers quantitative sequence analysis and, consequently, the establishment of structure–property relationships.

In the present work, we filled this gap using natural-abundance 2D ^{13}C–^{13}C INADEQUATE experiments, leveraging high-temperature cryoprobe NMR technology. Quantitative comonomer sequence analysis at triad level on custom-made copolymers prepared with different catalysts (**C1** and **C2** of Fig. 1) enabled determination of monomer reactivity ratios, and investigation of the impact of sequence distribution on molecular chain mobility, as reflected by the thermal behavior of E/V copolymers.

2. Experimental section

E/V copolymerization experiments were carried out in toluene at 80 °C using a high-throughput experimentation platform (Freeslate Parallel Pressure Reactor, PPR48), following previously described protocols [35,36]. Detailed experimental conditions are reported in the Supporting Information (Table S1).

All NMR experiments were performed at 125 °C in tetrachloroethane-1,2-d_2 with a Bruker Avance 400 spectrometer (400 MHz for ^{1}H) equipped with a 5 mm OD cryoprobe. Operating conditions for ^{13}C NMR were as follows: 45° pulse; 2.8 s acquisition time; 5.0 s relaxation delay; 1 K transients. Under these conditions, comparison with spectra acquired using inverse-gated decoupling confirmed that the nuclear Overhauser enhancement was comparable for all carbon resonances used for quantitative analysis. Therefore, spectra were acquired with continuous Waltz-16-modulated decoupling. Statistical analyses of dyad and triad distributions were carried out with the Copolstat software code [37]. 2D ^{13}C–^{13}C INADEQUATE experiments were executed with an acquisition time of 0.41 s and a pulse delay of 3.0 s; 400-800 transients per increment were acquired in a matrix of 2K ($t2$) × 516 ($t1$) data points, using the Waltz-64 decoupling sequence [38] and Bourdonneau's method for phase cycling [39]. The data were processed with shifted sine bell weighting functions and zero filling to form 4K ($t2$) × 1K ($t1$) data points. Acquisition time ranged from 48 to 72 h.

Differential Scanning Calorimetry (DSC) measurements were run with a DSC-1 setup by Mettler Toledo, operating between −100 °C and 180 °C at a scanning rate of 10 °C min^{-1} in a N_2 flow.

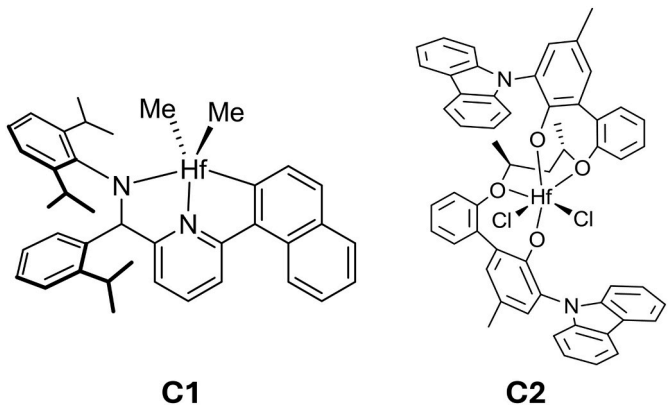


Fig. 1. Structures of the two non-metallocene (pre)catalysts used in this study.

3. Results and discussion

The literature ^{13}C NMR spectrum of an E/V copolymer sample featuring $x_V = 30.3\%$ (x_V = mole fraction of V units), with explicit resonance attributions, is shown in Fig. 2 [28]. Main-chain C atoms are labeled according to the nomenclature introduced by Carman and Wilkes [40], while C atoms of the cyclohexyl side branch are denoted as nB (B = Branch; n = 1-4, to indicate the number of bonds from the main-chain). The very weak peak at δ ~33.7 ppm assigned to methylene $S_{\alpha\alpha}$ is the only evidence of consecutive V units, evidently present in trace amounts. The peak at δ ~41.7 ppm was attributed to methines $T_{\beta\delta}$ and $T_{\delta\delta}$, which is inconsistent with the well-established additivity rules originally introduced by Grant and Paul [41] and subsequently refined by Lindeman and Adams [42], because said carbon atoms differ by at least one γ-gauche effect (corresponding to a chemical shift difference of approximately 2–3 ppm); this implies that the assignment in Fig. 2 requires revision.

The most reliable approach for resolving these ambiguities is represented by 2D^{13}C–^{13}C INADEQUATE (Incredible Natural Abundance DoublE QUAntum Transfer Experiment) experiments [43–46], which yield one-bond ^{13}C–^{13}C connectivities through J_{CC} couplings [43,47]. For samples at natural ^{13}C abundance this spectroscopy has an extremely poor sensitivity, which historically limited application; however, the introduction of NMR spectrometers with modern high-temperature cryogenic probes [44,48] has reduced the experiment time to practical values. Fig. 3 shows the 2D^{13}C–^{13}C INADEQUATE NMR map (48 h acquisition time) of an E/V copolymer prepared with catalyst system **C1**/MAO/BHT (Sample 1; MAO = methylalumoxane; BHT = butylated hydroxytoluene); the sample features $x_V = 17.9\%$ and almost exclusively isolated V units, like that of Fig. 2.

In the full ^{13}C–^{13}C connectivity map (Fig. 3-left), the signal at $\delta = 43.7$ ppm can be assigned unambiguously to methine $T_{\delta\delta}$, in view of the direct one-bond correlations with methylene $S_{\alpha\delta}$ ($\delta = 31.5$ ppm) and methine 4B ($\delta = 41.3$ ppm); this in turn directly correlates with methylenes 3B ($\delta = 30.4$ ppm). In the expanded region of the map (Fig. 3-right), the latter signal is clearly resolved from that at $\delta = 30.3$ ppm, which correlates with methylene $S_{\beta\delta}$ ($\delta = 28.1$ ppm) and therefore can be attributed to methylene $S_{\gamma\delta}$.

A second E/V copolymer (Sample 2) with $x_V = 44.1\%$ was prepared with catalyst system **C2**/MAO/BHT, which is able to homopolymerize V; the 2D ^{13}C–^{13}C INADEQUATE NMR map (72 h acquisition time) is shown in Fig. 4. Because of the high density of signals and the partial overlap of several cross-peaks in the full spectrum, magnified views of

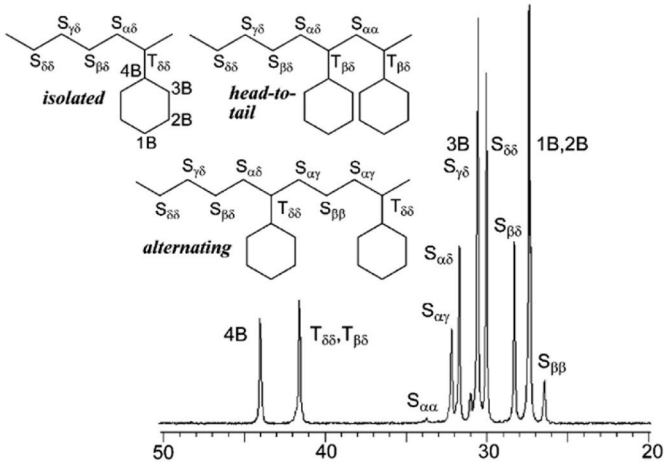


Fig. 2. ^{13}C NMR spectrum of an E/V copolymer sample ($x_V = 30.3\%$), with explicit resonance attributions (reproduced with permission from Ref. [28]. Copyright ©2005 American Chemical Society). Chemical shift values refer to HMDSO ($\delta = 0$).

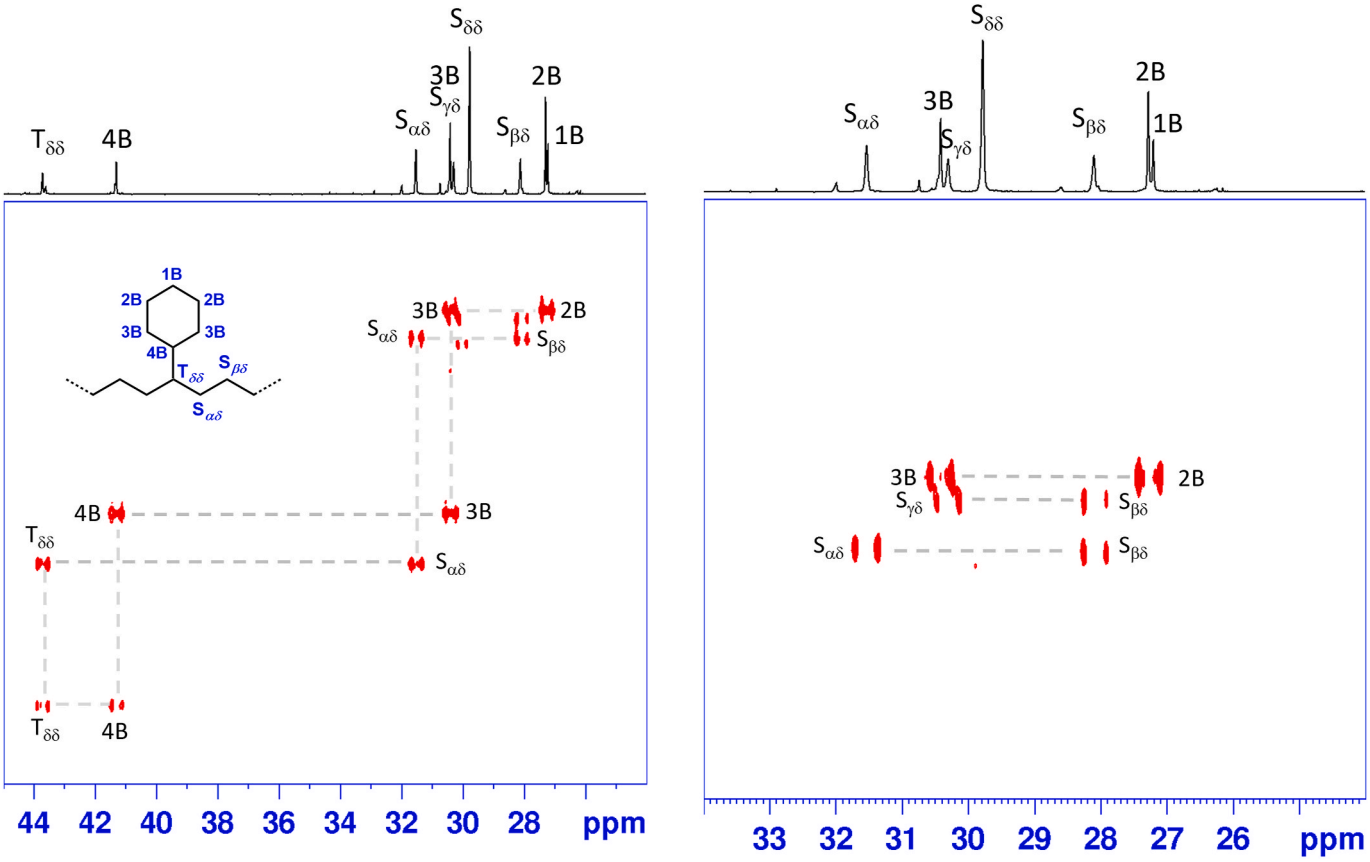


Fig. 3. (Left) 2D ^{13}C–^{13}C INADEQUATE NMR correlation map of Sample 1 ($x_V = 17.9\%$). (Right) Detail of the $\delta = 24$-34 ppm region. Chemical shift values refer to TMS ($\delta = 0$). Signal attributions and ^{13}C–^{13}C connectivities are indicated explicitly.

the most diagnostic regions are included in Fig. 4 to facilitate inspection of the key one-bond ^{13}C–^{13}C correlations supporting the assignments. The corresponding one-dimensional ^{13}C NMR projection exhibits a rich fine structure, reflecting a comonomer sequence distribution with V units in isolated (… EVE …), alternated (… EVEVE …), and consecutive (… E(V)$_n$E …, n ≥ 2) sequences; the latter are characterized by idiosyncratic correlation patterns, absent in the map of Sample 1. Although configurational effects associated with tacticity may in principle contribute to ^{13}C NMR signal splitting, particularly in V-rich copolymers containing VV and VVV sequences, they are not expected to represent the dominant origin of the fine structure observed here. The catalyst system used in this study is known to be highly stereoselective in α-olefin polymerization [30], and the corresponding polyvinylcyclohexane sample prepared with the same catalyst showed no significant evidence of stereochemical irregularities (Fig. S1). Accordingly, the main signal splittings are attributed primarily to constitutional sequence effects, although a very minor configurational contribution cannot be completely excluded. The multiplet at $\delta = 33.3$–33.8 ppm, barely visible in Fig. 2-Fig. 3 and correctly attributed to methylene $S_{\alpha\alpha}$, shows clear one-bond correlations with those at $\delta = 38.8$-39.4 ppm and $\delta = 40.6$-41.5 ppm, which can be assigned to methines $T_{\beta\beta}$ and $T_{\beta\delta}$, respectively; the separation of ~5 ppm between said signals is fully consistent with the summed contributions of two γ-gauche effects, in agreement with the aforementioned additivity rules [42].

Based on our comprehensive ^{13}C NMR assignment, summarized in Table 1, we calculated the full dyad and triad distributions for Samples 1 and 2 according to the method of Kakugo et al. [49] (see Supporting Information for details). The results are reported in Table 2, along with best-fit ones in the hypothesis of a first-order Markov copolymerization statistics [50]. The conditional probabilities directly obtained from the experimental dyad fractions, namely $p_{EE} = [EE]/[E]$ and $p_{VV} = [VV]/[V]$, are consistent with those obtained from the statistical fitting, supporting the validity of the first-order Markov treatment adopted in this work. The good agreement between the two data sets, reflected by reduced χ^2 values close to unity, provides an independent strong validation of the assignment and confirms that the sequence distribution of the investigated E/V copolymers is adequately described by a first-order Markov model.

Sample composition can be calculated from the ^{13}C NMR data as follows:

$$x_V = [VVV] + [VVE] + [EVE] = [VV] + 0.5\,[VE]$$

$$x_E = 1 - x_V$$

Equivalent results were independently obtained by ^{1}H NMR spectroscopy (see Supporting Information and Table 3).

The conditional probabilities of monomer insertion p_{XY} in Table 2 can be used to calculate monomer reactivity ratios, $r_E = k_{EE}/k_{EV}$ and $r_V = k_{VV}/k_{VE}$ [49] (see Supporting Information for details). The resulting values for catalyst systems **C1**/MAO/BHT and **C2**/MAO/BHT, along with the product $r_E \times r_V$, are summarized in Table 4. Both catalyst systems exhibit comparable r_E values, indicating a similar propensity to insert either monomer after an ethylene-ended chain. In contrast, substantial differences are observed for r_V, which directly reflects the ability of the catalyst to sustain consecutive V insertions. Accordingly, the very low $r_E \times r_V$ value obtained for **C1**/MAO/BHT indicates a higher tendency toward cross-propagation relative to VV propagation, whereas the higher value obtained for **C2**/MAO/BHT reflects a greater tendency toward consecutive V insertions.

As is well known, copolymerization statistics strongly impacts on copolymer physical properties. To evaluate the specific impact of sequence distribution independently of overall composition, we used

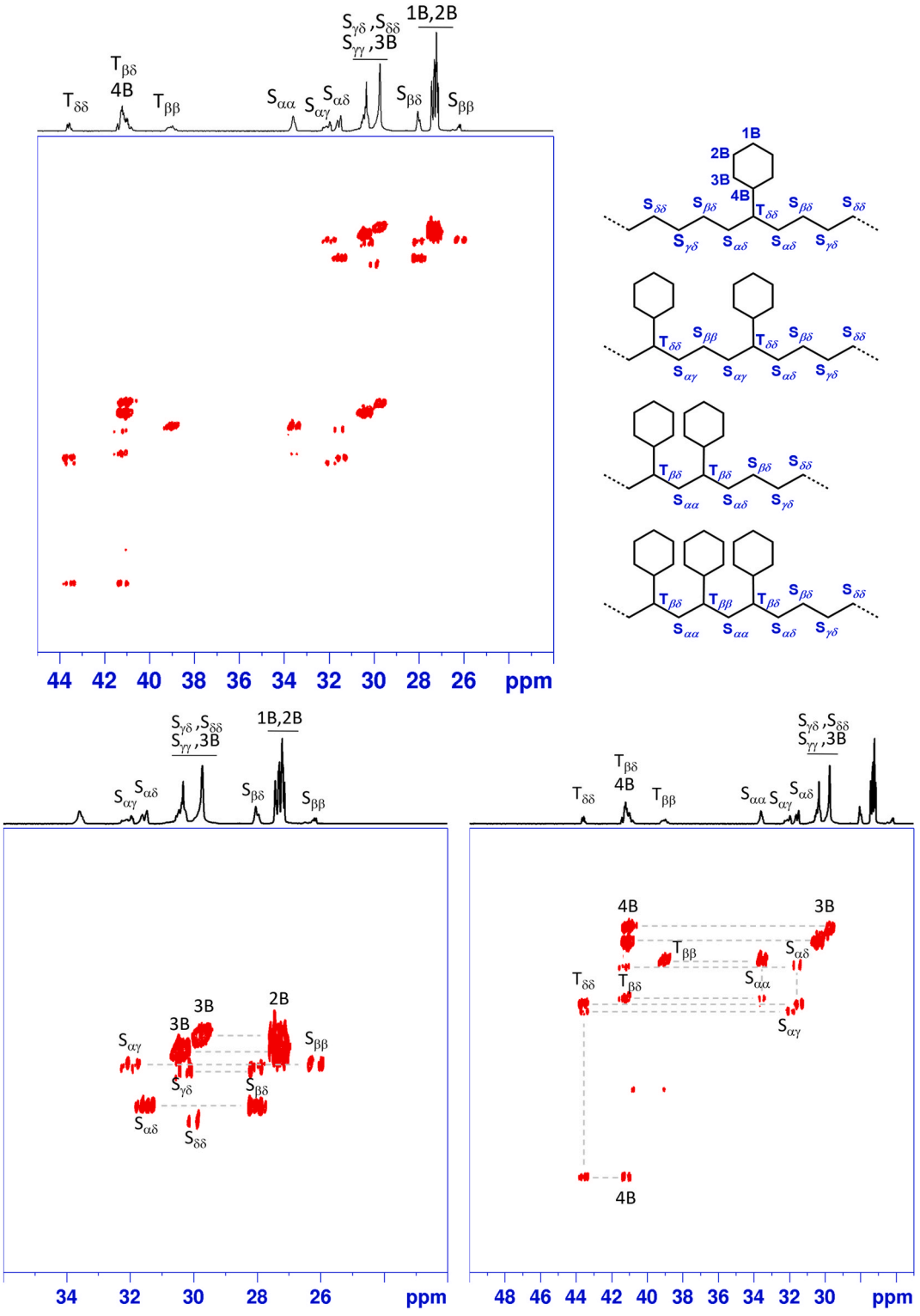


Fig. 4. (Top) 2D ^{13}C–^{13}C INADEQUATE NMR correlation map of Sample 2 ($x_V = 44.1\%$). (Bottom) Details of the map. Signal attributions and ^{13}C–^{13}C connectivities are indicated explicitly. Relative chemical shifts are referred to TMS ($\delta = 0$).

system **C2**/MAO/BHT to prepare a copolymer sample with x_v close to Sample 1 (Sample 3, $x_V = 18.1$ mol%). The results of ^{13}C NMR characterization and statistical analysis for Sample 3 are reported in the last two columns of Table 2, whereas the values of reactivity ratios and products thereof are shown in Table 4. It should be noted that the reactivity ratios independently derived from Samples 2 and 3 are in good agreement within experimental uncertainty (Table 4), confirming that the values obtained for **C2**/MAO/BHT are not significantly affected by the initial monomer feed composition and supporting the reliability of the first-order Markov analysis.

Because $r_E \times r_V$ is a kinetic descriptor and does not by itself fully describe the actual sequence distribution, especially in ethylene-rich copolymers, sequence parameters directly derived from the NMR dyad and triad distributions were also considered. In particular, the average

Table 1
^{13}C NMR resonance assignment for regioregular E/V copolymers (see text).

Carbon ID	$\delta(^{13}C)$, ppm (Sample 1)	$\delta(^{13}C)$, ppm (Sample 2)	Spectral Region in Fig. S1
$T_{\delta\delta}$	43.65	43.4-43.8	A
4B	41.28	40.6-41.5	B
$T_{\beta\delta}$	n.d.		
$T_{\beta\beta}$	n.d.	38.8-39.4	C
$S_{\alpha\alpha}$	n.d.	33.3-33.8	D
$S_{\alpha\gamma}$	n.d.	31.8-32.4	E
$S_{\alpha\delta}$	31.46	31.3-31.8	F
$S_{\gamma\gamma}$	n.d.	29.2-30.8	G
3B	30.35		
$S_{\gamma\delta}$	30.24		
$S_{\delta\delta}$	29.71		
$S_{\beta\delta}$	28.03	27.8-28.2	H
2B	27.21	26.9-27.6	I
1B	27.13		
$S_{\beta\beta}$	n.d.	26.0-26.4	L

sequence lengths n_E and n_V and the degree of randomness B [51], reported in Table 4, provide a more direct description of the copolymer microstructure. Sample 1 is characterized by $B = 1.2$, $n_E = 4.68$ and $n_v = 1.03$, indicating an alternation-enriched sequence distribution in which V units are predominantly isolated. By contrast, Sample 3, which has a comparable V content but was prepared with **C2**/MAO/BHT, shows $B = 0.9$, $n_V = 1.37$, and $n_E = 6.17$, indicating a sequence distribution slightly more blocky than random, with a higher probability of short V-containing sequences and longer ethylene-rich sequences. The different sequence distributions of Samples 1 and 3 are reflected in their thermal behavior (Fig. 5).

Although the two copolymers have comparable V contents and molecular weight ($M_n = 268$ kDa for Sample 1 and 401 kDa for Sample 3) and are both largely amorphous, the alternation-enriched Sample 1 shows lower values of $T_g \sim -23\,^\circ$C, $T_m \sim 23\,^\circ$C, and Δh_m ($\sim 6\,J\,g^{-1}$) compared with Sample 3 ($T_g \sim -14\,^\circ$C, $T_m \sim 45\,^\circ$C, and $\Delta h_m \sim 13\,J\,g^{-1}$). In contrast, the blockier sequence distribution of Sample 3 promotes the formation of longer ethylene-rich sequences capable of crystallizing more efficiently, while the presence of consecutive V units increases local chain rigidity and reduces segmental mobility. The increase in T_g of approximately 10 °C indicates that even very short V-rich sequences, present in low amounts, contribute to reduced segmental mobility of E/V copolymer chains, likely due to the sterically demanding and rather rigid cyclohexyl side branches. It is worthy to note that ethylene/1-hexene (E/H) copolymers made with catalyst system **C2**/MAO and featuring a composition close to Sample 3 ($x_H \sim 18.2\%$) exhibit a similar crystallinity ($\Delta h_m \sim 38\,J\,g^{-1}$) but a value of T_g well below −37 °C (see Supporting Information, Fig. S6). Since the sequence descriptors of this E/H copolymer (Tables S2–S3) are close to those obtained for Sample 3, the markedly lower T_g of the E/H copolymer is consistent with the distinct nature of the side chain. This comparison further supports the critical role of side-chain structure in governing molecular mobility and thermal behavior in ethylene-based copolymers.

Table 3
Values of x_V for Samples 1 and 2, as determined by means of ^{1}H NMR and ^{13}C NMR.

Sample	x_V (^{1}H NMR), %	x_V (^{13}C NMR), %
1	17.8	17.9 ± 0.1
2	44.0	44.1 ± 0.3

4. Conclusion

Despite the maturity of the polyolefin industry, the fine modulation of polymer chain microstructure enabled by coordination catalysis and the introduction of nonconventional monomers sustain an endless stream of novel materials with valuable properties. In this context, reliable microstructural elucidation by means of ^{13}C NMR spectroscopy represents a prerequisite for the implementation of such materials.

In this paper, we presented the case of ethylene/vinylcyclohexane (E/V) copolymers. By exploiting natural-abundance 2D ^{13}C–^{13}C INADEQUATE experiments enabled by high-temperature cryoprobe technology, we established a comprehensive and internally consistent ^{13}C NMR assignment of E/V copolymers, including for the first time resonances associated with consecutive V units. On this basis, quantitative dyad and triad sequence distributions were determined, enabling the determination of monomer reactivity ratios and a detailed description of copolymerization statistics. Although the two catalytic systems investigated display comparable insertion behavior with respect to ethylene units, they differ markedly in their ability to sustain consecutive V insertions, as captured by the $r_E \times r_V$ product. The sequence parameters derived from the NMR dyad and triad distributions show that the **C1**-based system produces an alternation-enriched sequence distribution, with V units predominantly isolated within ethylene sequences, whereas the **C2**-based system gives a less alternating distribution with higher

Table 4
Monomer reactivity ratios and product thereof for catalyst systems **C1**/MAO/BHT and **C2**/MAO/BHT, along with sequence parameters for Samples 1-3.

Sample	Precatalyst	r_E	r_V	$r_E \times r_V$	n_E[a]	n_V[a]	B[b]
1	**C1**	19	0.003	0.07	4.68	1.03	1.2
2	**C2**	14	0.14	1.8	2.64	2.12	0.8
3	**C2**	16	0.11	1.8	6.17	1.37	0.9

[a] Number-average sequence length.
[b] randomness parameter [51].

Table 2
Experimental ^{13}C NMR dyad and triad distributions for Samples 1-3, and best-fit calculated ones assuming a first-order Markov copolymerization statistics (see text).[(a)]

Sequence[a]	Sample 1		Sample 2		Sample 3	
	Exp., %	Calc., %	Exp., %	Calc., %	Exp., %	Calc., %
[EE]	64.4 (3)	64.73	34.8 (5)	34.91	68.5 (5)	68.75
[EV]	35.1 (3)	34.97	41.9 (5)	41.71	26.5 (5)	26.69
[VV]	0.5 (3)	0.30	23.6 (5)	23.38	5.0 (5)	4.56
[EEE]	51.2 (3)	50.96	22.7 (10)	21.85	58.1 (10)	57.57
[EEV]	26.8 (3)	27.53	24.7 (10)	26.11	21.0 (10)	22.35
[VEV]	4.2 (3)	3.72	9.0 (5)	7.80	2.9 (5)	2.17
[EVE]	17.3 (3)	17.18	10.0 (5)	9.83	10.3 (5)	9.94
[EVV]	0.4 (3)	0.60	20.4 (5)	22.05	6.1 (5)	6.80
[VVV]	n.d.	0.01	13.0 (5)	12.36	1.6 (5)	1.16
		$p_{EE} = 0.787$		$p_{EE} = 0.626$		$p_{EE} = 0.837$
		$p_{VV} = 0.017$		$p_{VV} = 0.529$		$p_{VV} = 0.255$
		$\chi^2_r = 1.4$		$\chi^2_r = 2.6$		$\chi^2_r = 1.1$

[a] p_{XY} = conditional probability of inserting comonomer Y after comonomer X (X and Y = E or V).

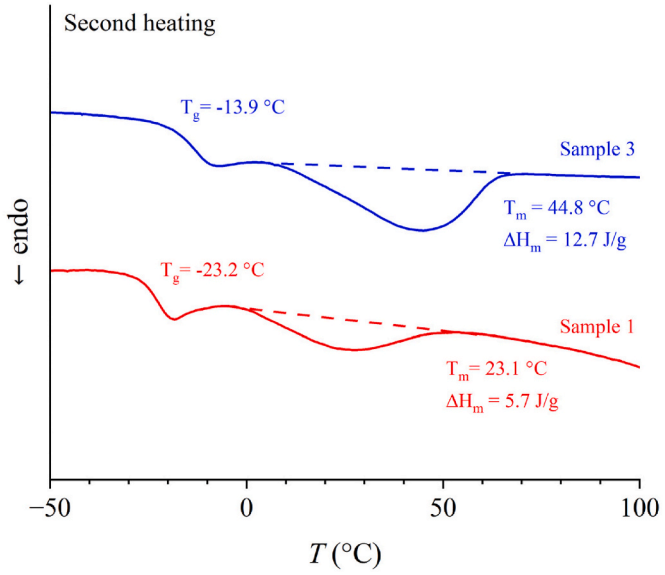


Fig. 5. DSC traces (2nd heating) of Sample 1 (red trace) and Sample 3 (blue trace).

monomer persistence.

Importantly, these microstructural differences translate into measurable variations in macroscopic properties. Copolymers with comparable composition and molecular weight but different sequence distributions exhibit significant differences in glass-transition temperature ($\Delta T_g \approx 10\,^{\circ}C$), indicating that sequence distribution can be exploited as an additional design variable for tailoring the thermal behavior of E/V copolymers.

The approach described here opens the door to quantitative studies correlating catalyst structure, product microstructure and materials properties for this class of innovative polyolefin products, thereby enabling a more rational design of advanced polyolefin materials.

CRediT authorship contribution statement

Roberta Cipullo: Conceptualization, Formal analysis, Investigation, Methodology, Supervision, Validation, Writing – original draft, Writing – review & editing. **Lucrezia Ricciardi:** Data curation, Formal analysis, Investigation, Visualization, Writing – review & editing. **Viviana Losciale:** Data curation, Formal analysis, Investigation, Validation, Writing – review & editing. **Andrea Pucciarelli:** Formal analysis, Investigation, Writing – review & editing. **Antonio Vittoria:** Formal analysis, Investigation, Writing – review & editing. **Vincenzo Busico:** Conceptualization, Methodology, Resources, Supervision, Validation, Writing – original draft, Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.polymer.2026.130510.

Data availability

Data will be made available on request.

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